

North Utilities Operator Rounds and Alarm Notes

Equipment: CWP-3, CT-2, seal-flush panel, and condenser-water loop | Evidence window: 2026-05-04 to 2026-05-18

Use note: operator observations support alarm timing, visible leakage, unusual noise, and sequencing. Instrumented values in BAS trends, route logs, field readings, inventory, and OEM criteria control numerical values.

source_id	timestamp	area	operator note	alarm or observation	control use
OP-001	2026-05-04 05:36	North pump gallery	CWP-3 was running but header flow on HMI looked lower than usual. Suction valve handle appeared open.	LF-203 low-flow alarm	sequence only
OP-002	2026-05-04 05:50	Chiller room	Chiller 1 unload command appeared after flow alarm. No unusual noise at chiller.	load shed after LF alarm	sequence only
OP-003	2026-05-06 07:18	Pump gallery	Pump sounded gravelly near suction spool. No local strainer DP indicator visible.	suction noise	supports strainer check
OP-004	2026-05-07 15:45	Pump gallery	After coupling spider change, pump noise was lower but vibration alarm did not clear from screen.	vibration alarm persisted	sequence only
OP-005	2026-05-08 13:55	BAS workstation	Temperature point labelled CT-2 fan bearing showed high value. Route tech said sensor was physically on CWP-3 pump bearing.	label confusion	supports tag conflict
OP-006	2026-05-09 04:54	Pump gallery	Seal guard drain showed wet streaks and intermittent drip. No flush pressure gauge on panel.	visible leakage	supports seal leak and flush gap
OP-007	2026-05-09 05:03	Chiller room	Chiller 1 tripped and reset after condenser-water flow alarm. CT-2 fan was running.	short trip	impact timing
OP-008	2026-05-10 18:00	Pump gallery	Maintenance told operations the seal job was closed. Mist was still visible at guard drain during restart.	mist after close-out	supports disputed seal repair
OP-009	2026-05-11 16:25	Seal-flush panel	New local pressure gauge showed about 19 psi while pump was running. Seal gland still wet.	low flush pressure	observation only; gauge reading corroborated by WO
OP-010	2026-05-12 07:50	Cooling tower roof	CT-2 basin screen was matted. Tower approach on local board looked high. Pump seal still wet in gallery.	tower debris and seal leak	separate symptoms
OP-011	2026-05-12 12:15	Cooling tower roof	After screen cleaning, tower approach looked normal. Pump vibration alarm still present in BAS summary.	approach improved	supports tower not root of vibration
OP-012	2026-05-15 15:25	Pump gallery	After strainer clean, flow improved and pump sounded less gravelly. Vibration alarm remained active.	flow improved, vibration remained	supports strainer contributor only
OP-013	2026-05-17 06:12	Chiller room	Chiller 1 tripped again during low-flow alarm. Tower temperatures looked normal on local panel.	trip with normal tower approach	supports pump flow issue
OP-014	2026-05-18 10:25	Pump gallery	Flush filter was changed and gauge pressure improved. Seal gland still had wet ring after run.	leak not cleared	supports seal unresolved

Alarm Chronology Extract

source_id	timestamp	alarm tag	duration	operator action	note
ALM-001	2026-05-04 05:35	LF-203 CWP low flow	18 min	load shed accepted	First low-flow event in window. Strainer DP not available.
ALM-002	2026-05-07 15:38	VIB-203 pump vibration	persistent	called maintenance	Coupling spider replacement did not clear alarm.
ALM-003	2026-05-09 05:02	CH1-COND-TRIP	6 min	reset chiller after flow stabilized	Seal leak visible before trip.
ALM-004	2026-05-10 18:08	SEAL-LEAK observation	not timed	noted in log	Mist remained after CMMS seal close-out.
ALM-005	2026-05-12 10:14	TEMP-203 bearing temperature	22 min	reduced load	BAS label still displayed tower fan bearing.
ALM-006	2026-05-12 10:18	APPROACH-CT2 high approach	47 min	sent tower check	Basin screen matted; cleaned later.
ALM-007	2026-05-15 14:03	STR-DP action	45 min	called mechanic for strainer clean	Temporary DP taps showed high differential.
ALM-008	2026-05-17 06:10	LF-203 CWP low flow	27 min	reset chiller sequence	Tower approach normal; BAS flow scaled correctly.
ALM-009	2026-05-18 11:48	CH1-COND-TRIP	5 min	reset after flow recovered	Strainer DP no longer at action level.

Operator interpretation limits

OP-LIMIT-001: Operator notes cannot establish measured vibration, flow, pressure, alignment, temperature, or inventory status when instrumented records exist.

OP-LIMIT-002: Observations of visible leakage, noise, alarm sequence, and tower cleaning are valid corroborating evidence for timing and symptom location.

OP-LIMIT-003: HMI flow comments before 2026-05-15 require scaling reconciliation because the BAS point used a different range than the loop card.

Shift Handover Notes

source_id	shift	handover text	risk tag
SH-001	2026-05-10 day to night	Maintenance close-out says seal replaced, but night operator should watch guard drain because mist was still present during ramp-up.	seal conflict
SH-002	2026-05-11 day to night	Alignment crew completed laser work. Pump still has vibration alarm. No note showing suction support was checked after final reading.	alignment conflict
SH-003	2026-05-12 day to night	Tower basin was cleaned and approach improved. Do not assume this fixed pump vibration or seal leakage.	tower contributor
SH-004	2026-05-15 day to night	Flow transmitter scaling was corrected in BAS. Use values after 10:20 as current flow basis. Earlier BAS values need correction.	flow scaling
SH-005	2026-05-18 day to night	Flush filter changed and pressure improved. Seal ring remains wet. Correct seal replacement still not confirmed in stores history.	seal unresolved

End of operator notes. Source IDs OP-001 through OP-014, ALM-001 through ALM-009, and SH-001 through SH-005 are the only operator note source IDs in this file.